Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	15 February 2026
Team ID	LTVIP2026TMIDS54324
Project Name	Smart Sorting: Identifying Rotten Fruits and Vegetables Using Transfer Learning
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	To User	Introduction about Smart Sort App
		Instructions about how to use
FR-2	Image Upload / Input	Upload image of fruits/vegetables
		Capture image via camera
FR-3	Prediction / Smart Sorting	Identify rotten vs fresh produce using transfer learning
		Provide confidence score for prediction
		Suggest sorting action (e.g., discard / keep)

FR-4	**View Results / Reports**	**Display classification result immediately**
		Show past predictions history (optional)
FR-5	**Admin / Dataset Management (if applicable)**	**Upload new training data (admin)**
		Trigger model retraining (admin)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	**Non-Functional Requirement**	**Description**
NFR-1	**Usability**	The system should have a clean, intuitive UI for users to easily upload images and view results without technical expertise.
NFR-2	**Security**	The system should protect user data (images, login info) using encryption and secure authentication methods.
NFR-3	**Reliability**	The system should consistently provide accurate predictions with minimal failure or downtime during usage.

NFR-4	**Performance**	The prediction response time should be under 2 seconds for a single image classification.
NFR-5	**Availability**	The system should be available 24/7 with minimal service interruptions.
NFR-6	**Scalability**	The solution should handle increasing users or image inputs by scaling the model inference service and storage infrastructure as needed.